

Solution
HUMAN EYE -02
Class 10 - Science
Section A

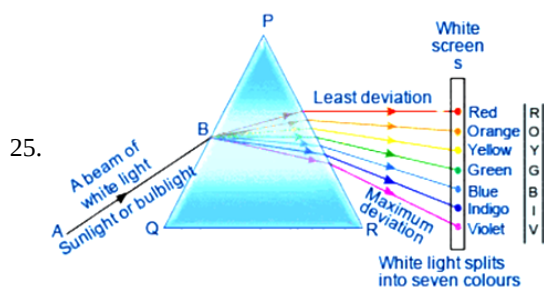
1.
(d) $\angle A$ and $\angle e$
Explanation: $\angle A$ and $\angle e$
2.
(b) $\angle r$, $\angle A$ and $\angle D$
Explanation: $\angle r$, $\angle A$ and $\angle D$
3.
(d) $\angle A$ and $\angle i$
Explanation: $\angle A$ and $\angle i$
4. (a) p, y, z
Explanation: p, y, z
5. (a) $\angle 2 + \angle 3 = \angle 1 + \angle 4$
Explanation: Only this option is correct

Section B

6. (a) - (iv), (b) - (iii), (c) - (ii), (d) - (i)
7. (a) - (iv), (b) - (iii), (c) - (ii), (d) - (i)
8. (a) - (iv), (b) - (iii), (c) - (ii), (d) - (i)
9. (a) - (iv), (b) - (iii), (c) - (ii), (d) - (i)
10. (a) - (iv), (b) - (iii), (c) - (ii), (d) - (i)
11. (a) - (iv), (b) - (iii), (c) - (ii), (d) - (i)
12. (a) - (iv), (b) - (iii), (c) - (ii), (d) - (i)
13. (a) - (iv), (b) - (iii), (c) - (ii), (d) - (i)
14. (a) - (iv), (b) - (iii), (c) - (ii), (d) - (i)
15. (a) - (iv), (b) - (iii), (c) - (ii), (d) - (i)
16. (a) - (iv), (b) - (iii), (c) - (ii), (d) - (i)
17. (a) - (iv), (b) - (iii), (c) - (ii), (d) - (i)
18. (a) - (iv), (b) - (iii), (c) - (ii), (d) - (i)
19. (a) - (iv), (b) - (iii), (c) - (ii), (d) - (i)

Section C

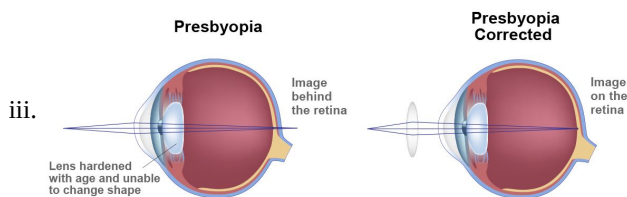
20. Power, $P = -0.5D$
Let x be the distance of the far point.
The focal length, $f = -x$
Power of a lens is given by
$$P = \frac{1}{f(\text{ in metres })}$$
$$-0.5 = \frac{1}{-x}$$
$$-x = \frac{1}{0.5} = +2 \text{ m}$$
21. When the beam of light is redirected in different direction by interaction of the particles present in the medium it is called scattering of light.
A clear cloudless day-time sky is blue because molecules in the air scatter blue light from the sun more than they scatter red light.
22. Draw the boundary of prism on the drawing sheet.
It is done so as to readjust the prism to same position if it gets displaced while doing experiment.
23. Refraction of light.
24. The lens becomes thick and the ciliary muscles contract when a person looks at nearby objects. Hence, focal length of the lens decreases.



Dispersion of light

Cause: The dispersion of white light occurs because colors of white light travel at different speeds through glass prism. Different colours undergo different deviations on passing through prism.

26. Rainbow is a natural spectrum formed in the sky after rain shower due to dispersion of light caused by suspended water droplets in atmosphere.
27. The ability of the eye to focus objects lying at different distances clearly on the retina by changing the focal length of the eye lens with the help of ciliary muscles is called the power of accommodation of the eye.
28. The phenomenon responsible for increasing the apparent length of the day by 4 min is **atmospheric refraction**.
29. In the persons suffering from long sightedness, the image is formed at the back of retina.
30. The intensity of scattered light is directly proportional to the square of the amplitude (a) of scattered light, i.e. $I \propto a^2$ means greater the amplitude greater will be intensity of refraction and vice versa.
31. If the earth has no atmosphere, the sky would have appeared black because of no refraction.
32. As red light is scattered the least and it covers longer distances. That's why it is used in danger signal.
33. The light ray on emerging out of a prism always bends towards the base of the prism.
34. Tyndall effect is responsible for the visibility of the path of light in the dark room filled with smoke.
35. i. Along with age, the power of the accommodation factor to adjust the focal length also decreases. People have difficulties viewing nearby objects clearly without the assistance of corrective eyeglasses.
ii. Condition is known as presbyopia.



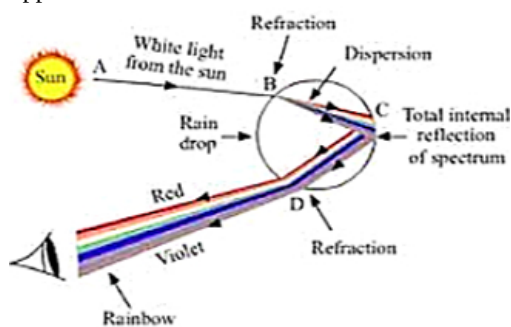
36. The component of white light which is least scattered by fog or smoke is **red color**.
37. The focal length is given by the reciprocal of power i.e.

$$\text{Focal Length} = \frac{1}{\text{Power of lens}}$$

$$\text{Focal length} = \frac{-100}{4.5} \text{ cm}$$

$$\text{Focal length} = -22.2 \text{ cm}$$
 The focal length of the corrective lens is -22.2 cm
38. Accommodation of the eye is the property of the eye responsible for the variable focal length of the eye lens.
39. $\angle A$ and $\angle e$ are correctly labelled. $\angle A$ is angle of prism whereas $\angle e$ is angle of emergence.
40. Infra-red is found at P and ultraviolet is found at Q.
41. The bees are able to see ultraviolet light because the retina of bees contains cone cells that are sensitive to ultraviolet light.
42. The sky appears blue because of the presence of the atmosphere which scatters blue light. Space has no atmosphere and no light is scattered, hence it appears dark.
43. i. Pupil
ii. Retina
44. Ciliary muscles help in changing the focal length of the eye lens.
45. When image formed at retina, light sensitive cells get activated and generate electrical signal. These signals are sent to brain via optic nerve.
46. Hot iron emits infrared rays. These rays are invisible to the eyes.
47. i. Myopia.
ii. Prolonged reading and other close-up eye activities; genetic factors; stress from the environment; etc.
iii. Concave lens are diverging lenses, which diverge upon seeing rays to meet them on the retina.

48. The function of vitreous humour is to provide the eye its form and shape.
49. It provides the fine adjustment of focal length of eye lens system so as to focus images of objects situated at different distances on the retina.
50. Myopia or short-sightedness. The concave lens should be used to correct his vision.
51. Size of the particles of the medium is the factor on which the color of the scattered light depends.
52. If on touching, the lens feels bulging out, it is a convex lens and the person is suffering from hypermetropia (long sightedness).
And if the lens is thinner from the center, it means it is a concave lens and the person is suffering from myopia (short sightedness).
53. As the focal length of glasses is positive, so the power of the lens is also positive which indicates the use of a convex lens. Hence, he is suffering from hypermetropia.
54. The red light has a wavelength of about 1.8 times greater than blue light. Thus, when sunlight passes through the atmosphere, the fine particles in the air scatter the blue color (shorter wavelengths) more strongly than red. The scattered blue light enters our eyes.
55. Wavelength of light.
56. i. In the upper part, a concave lens is used
ii. In the lower part, a convex lens is used.
57. For the persons suffering from near-sightedness, the image is formed before retina.
58. The negative sign of the power of lens indicates that the focal length of the lens used for correction is negative. This means that a concave lens is used for correction.
59. Water droplets acts as tiny prism in the sky. The sunlight when enters these tiny droplets undergo internal reflection and also refract these rays which are dispersed causing a band of seven colours called rainbow. Rainbow is always formed in the direction opposite to the sun.



60. Here,
The distance of far point, $x = 100$ cm
Focal length, $f = -x = -100$ cm
Using the formula, $P = \frac{100}{f}$
Substituting the values, we get $P = \frac{100}{-100} = -1$ D
The statement is wrong.
61. The negative sign of the power of lens indicates that the focal length of the lens used for correction is negative. A concave lens is used for the correction of myopia and hence, the person is suffering from myopia.
62. The object distance, $u = \infty$
Image distance, $v = -4$ m
Using the formula, $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$
We get $\frac{1}{f} = \frac{1}{-4} - \frac{1}{-\infty}$
 $\frac{1}{f} = -\frac{1}{4} + 0$
 $f = -4$ m
In order to calculate power, $P = \frac{1}{f(\text{ in metres })}$
 $P = \frac{1}{-4} = 0.25$ D
63. The red cloth will appear black in the blue portion of spectrum of light.
64. Cornea and Aqueous humour refract most of the light entering the eye.
65. This is because of scattering of light near the horizon, most of the blue light and shorter wavelengths are scattered away by the particles present in the atmosphere during sunrise and sunset. So, the light that reaches our eyes is of longer wavelength (e.g. red). This gives rise to the reddish appearance of the sky. But during the day sun appears white as sun is near the surface of earth nearly overhead, thus the sunlight passes through much smaller distance and thus the scattering is much less and sun appears white.

66. When we are in bright sunlight the aperture of the pupil would be small to regulate the amount of light entering the eye preventing glare, discomfort and damage to eyes. As we enter a dark room less amount of light would enter our eyes due to small size of pupil, and we won't be able to see objects clearly. It takes some time to regulate the size of the pupil through iris. Hence, it requires some time to see things.

67. 1. Functions of following parts of human eye are given below :

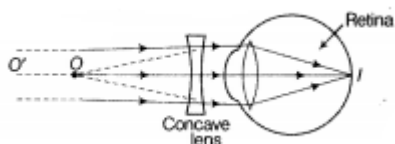
1. Cornea - It is a thin membrane which provides 67% of the eye's focussing power.
2. Iris - It controls amount of light entering the eye by controlling the size of pupil similar to the aperture of a camera which has capacity to decrease or increase the amount of light entering eye.
3. Crystalline lens - It helps to focus light on retina for image formation.
4. Ciliary muscles - It contracts and relax in order to change the lens shape for focussing image at retina. when it contracts the lens become thicker and when it relaxes the lens become flat.

2. 1. The objective of organising such campaigns is to guide, educate and help those people who are suffering from corneal blindness that they can be cured by corneal replacement surgery.
2. 1. Come to participate in this campaign because, if someone get his vision through your eyes, it is an incredible help.
2. As eye is one of the most valuable sense organs through which an individual can achieve so many things in his/her life, so try to realise the situation that these people are suffering from.
3. The persons who actively participate and contribute in such programme are strong hearted and very much helpful for the people living in such situations.

68. i. Near sightedness (myopia) defect arises either because of :

(a) decrease in focal length of eye lens.(b) elongation of the eye ball

ii. To correct this defect of vision, he must use a concave lens of suitable focal length. The concave lens of suitable focal length will bring the image back to the retina as shown in the given figure.



iii. Given, $v = -100 \text{ cm}$, $u = \infty$

Using lens formula,

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u} \Rightarrow \frac{1}{-100} - \frac{1}{\infty} = \frac{1}{f}$$

$$f = -100 \text{ cm} = -1 \text{ m.}$$

$\therefore$ Power of lens,

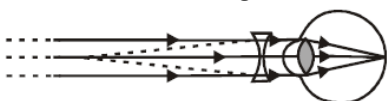
$$P = \frac{1}{f(m)} = \frac{1}{-1} = -1\text{D.}$$

69. A camera in many ways is similar to the human eye as both eye and camera has convex lens. But, there are some basic differences in image formation between the two as follows:

- i. In camera, the distance between the lens and the screen can be adjusted but not the focal length of the lens. However, in eye, the ciliary muscles adjust the focal length keeping the distance between the lens and the retina constant.
- ii. The image formed on the retina is temporary and its impression is recorded in the brain as memory. However, the image formed on the film of camera is a permanent record.

70. i. The eye suffering from myopia or short-sightedness, has long eyeball than that of normal eye due to which the retina is at a larger distance from the eye lens thus image formation occurs before retina rather than onto it.
- ii. The eye suffering from hypermetropia or long-sightedness has short eyeball than that of normal eye due to which the retina is at smaller distance from the eye lens thus, the formation of the image occurs behind the retina and not on retina.

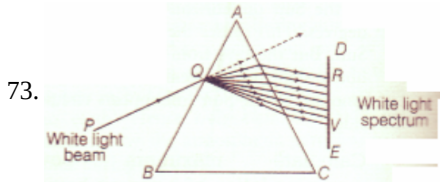
71. This student is unable to see far off objects. This means that the student is suffering from myopia. Doctor will prescribe a concave lens a suitable focal length.



Correction for myopia

72. Different wavelengths deviate differently in the prism because the angle of refraction for different colours having different wavelengths is different while passing through the glass prism (medium). A light ray is refracted when it passes from one medium to another at an angle and its speed changes. At the interface, it is bent in one direction if the material it enters is denser (when light slows down) and in the other direction if the material is less dense (when light speeds up). Because different wavelengths

(colours) of light travel through a medium at different speeds, the amount of bending is different for different wavelengths. Violet is bent the most and red the least because violet light has a shorter wavelength, and short wavelengths travel more slowly through a medium than longer ones do.



- i. The phenomenon of splitting of white light into its constituent colours is called dispersion of light. It is caused due to difference in speed of constituent colours of light travel in the medium other than air/vacuum because of different speed they bend at different angles.
 - ii. In nature, this Phenomenon is observed in formation of rainbow where all the seven colours constituting white light is visible.
 - iii. Based on phenomenon of dispersion, we can conclude that
 - a. White light consists of seven colours. Violet, indigo, blue, green, yellow, orange and red.
 - b. Violet light suffers maximum deviation and red light suffers minimum deviation.
74. There are mainly four common defect of vision that can be corrected by the use of suitable eyeglasses or spectacles. There are
- i. Myopia or near-sightedness,
 - ii. Hypermetropia or far-sightedness,
 - iii. Presbyopia, and
 - iv. Astigmatism